



DataDriven Floorplanning

ICCAD Problem C

B12901012 張維傑

B12901080 李俊翰

B12901185 謝承恩



Outline

- Recap
- Attention Model
- GNN Model
- Legalization



Recap: DataDriven Floorplanning

- Floorplanning with lots of constraints
- Given 1M data for training
- Input
 - Block areas
 - Constraints
- Output
 - (x, y, w, h) for each block



Recap: DataDriven Floorplanning

- Hard Constraints

- Block area -> within 1% error
- Overlap-free -> No overlap is allowed
- Fixed shape -> Aspect ratio is fixed
- Preplaced block -> (x, y, w, h) is fixed

- Soft Constraints

- Grouping -> some blocks need to be abutted
- MIB -> some blocks need to share the same (w, h)
- Boundary -> some blocks need to be placed on the corner or the side

$$\text{Cost} = \begin{cases} \min \left(\left(1 + \alpha \cdot (\text{HPWL}^{\text{gap}} + \text{Area}_{\text{bbox}}^{\text{gap}}) \right) \times e^{\beta \cdot \text{Violations}^{\text{relative}}} \times \max(0.7, \text{RuntimeFactor}^{\gamma}), M - 10^{-6} \right) \\ M, \end{cases}$$



Attention Model



- Normalization
- Connectivity Graph
- Padding & Masking

- Block Encoder
- Connectivity Biased Attention
- Denoiser -> MSE loss
- EMA
- **Soft constraints loss**

- DDIM
- Coordinate Transform
- Multi-solution
- **MIB force**



Config:

- 200 k Training Data
- 100 epochs

Attention Model - Training

- **Block Encoder:** Encode area & topology into **condition embeddings**
- **Connectivity Biased Attention:** $\text{softmax}(\frac{QK^T}{\sqrt{d_k}} + \text{conn_bias} + \text{type_bias})$ → Inject **physical connectivity** as prior knowledge
- **Denoiser:** Predict injected noise at random timestep t -> **MSE Loss**
- **EMA:** Maintain **shadow weights** for stable and higher-quality inference
- **Soft constraints loss:** Add soft constraints loss to help model training with the soft constraints
 - Weight: grouping: 0.3; boundary: 0.3; MIB: 0.05



Attention Model - Inference

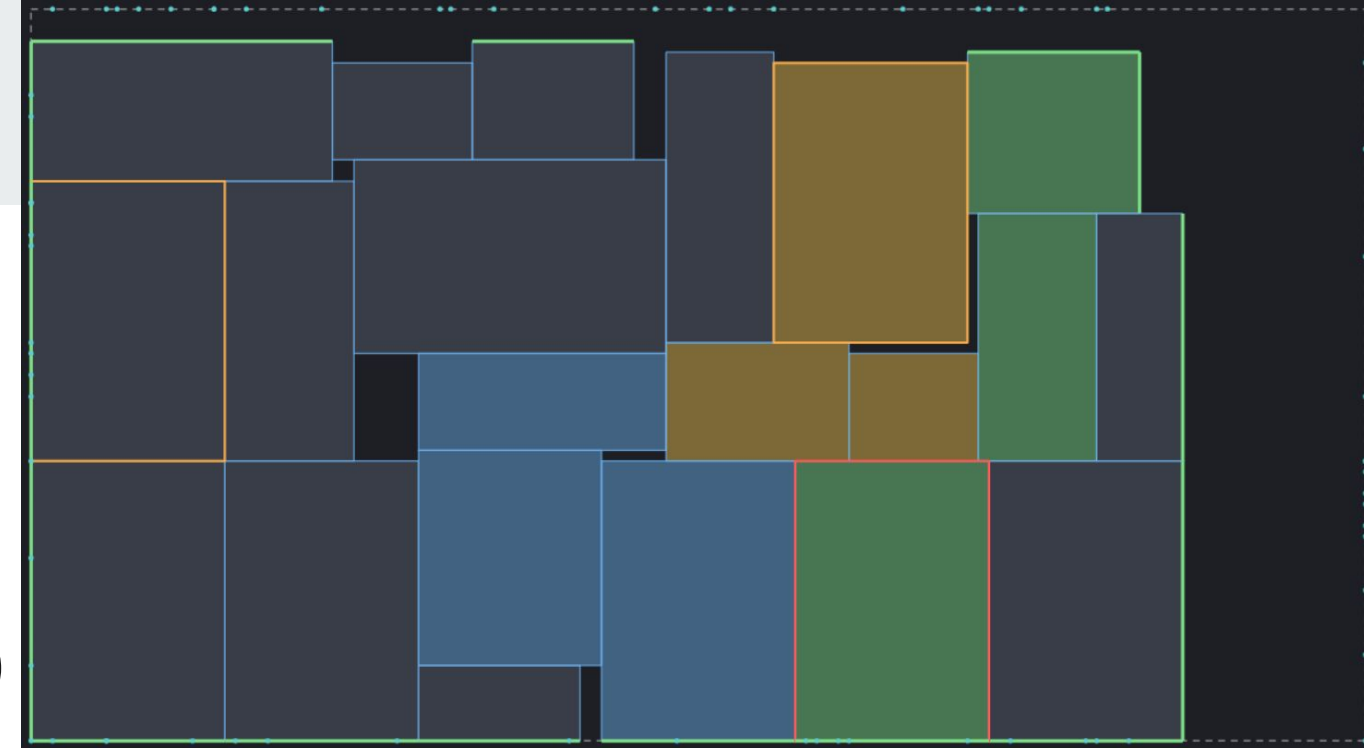
- **DDIM:** Accelerate sampling process (1000 steps -> 50 steps)
 - No noise adding when inference
 - Predict the final x_0 and aim to it
- **Coordinate Transform:** Transform the compressed coordinate back
- **Multi-solution:** Generate multi-solution and choose the **least overlap one**
- **MIB Force:** At the end of inference, force the MIB to have the same aspect ratio



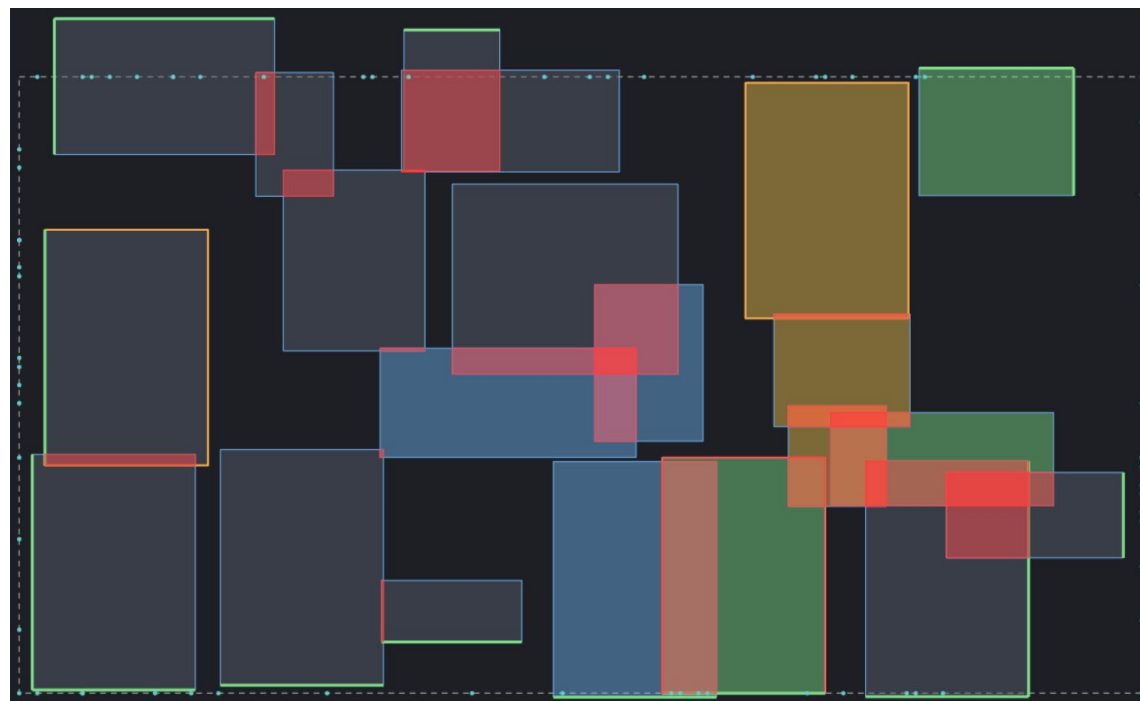
Attention Model - Deal with constraints

- **Fixed shape:** Fixed the aspect ratio when inferencing
- **Preplaced:** Fixed (x, y, w, h) when inferencing
- **Grouping:** Block encoder & training loss & attention_bias
- **MIB:** Block encoder & training loss & MIB force when inferencing
- **Boundary:** Block encoder (4 dimensions) & training loss

Attention Model - Result (21 blocks)

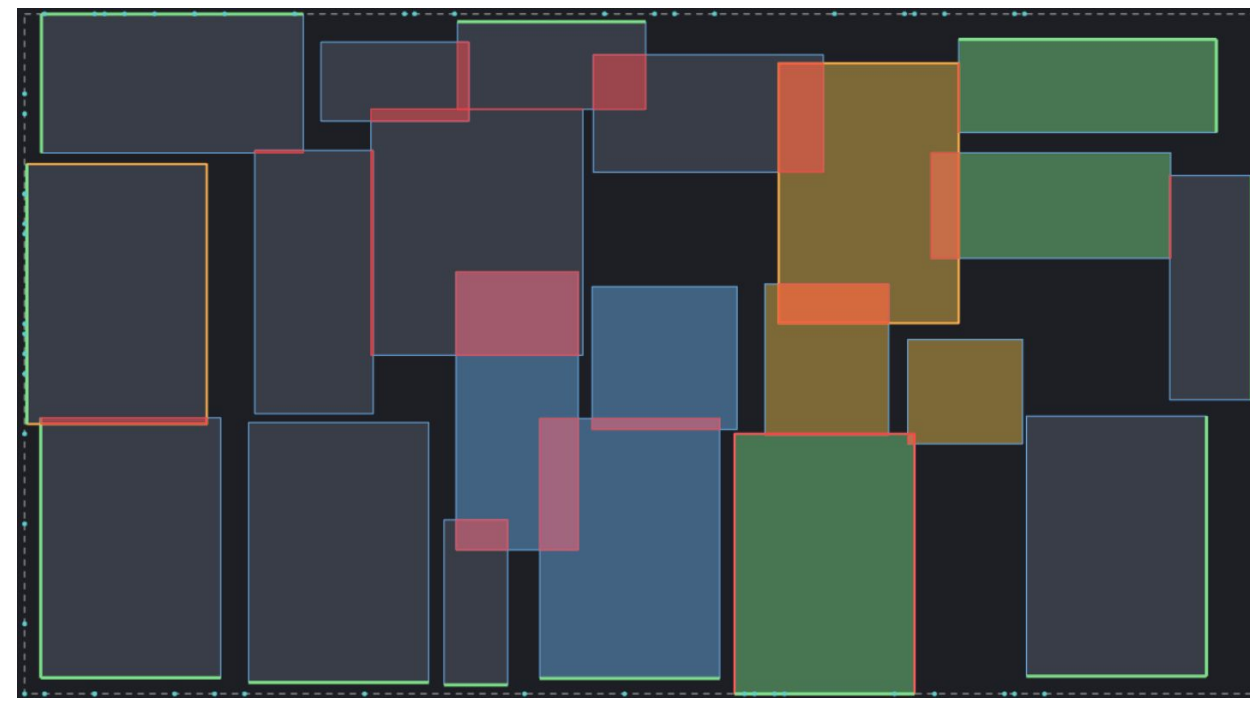


epoch 30



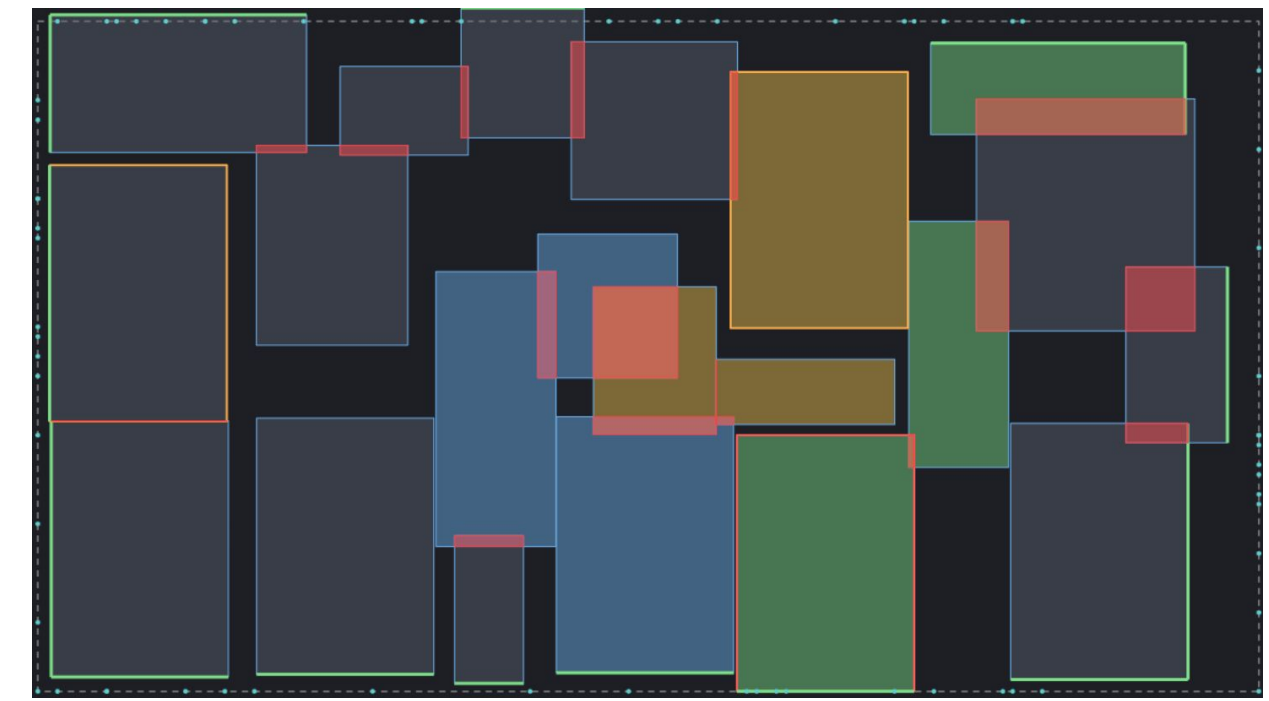
Area gap vs. optimal: +29.7%
HPWL gap vs. optimal: +11.4%
Overlap area: 885.1 / 9019.5
Soft constraints violation: 0.565
(b = 7, g = 6, m = 0)

epoch 60



Area gap vs. optimal: +19.7%
HPWL gap vs. optimal: +8.1%
Overlap area: 371.6 / 8328.1
Soft constraints violation: 0.478
(b = 5, g = 6, m = 0)

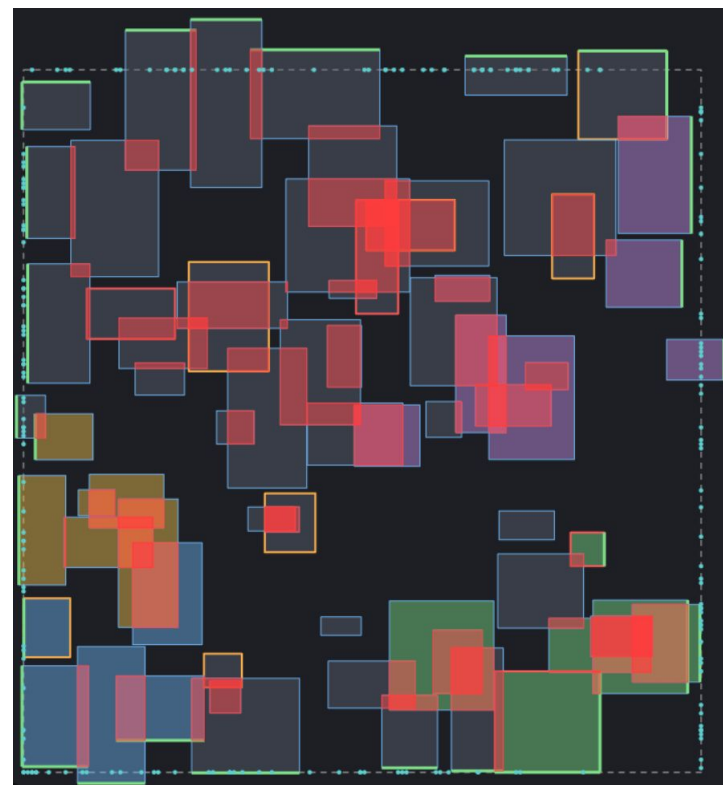
epoch 100



Area gap vs. optimal: +19.3%
HPWL gap vs. optimal: +7.0%
Overlap area: 341.8 / 8296.4
Soft constraints violation: 0.478
(b = 5, g = 6, m = 0)

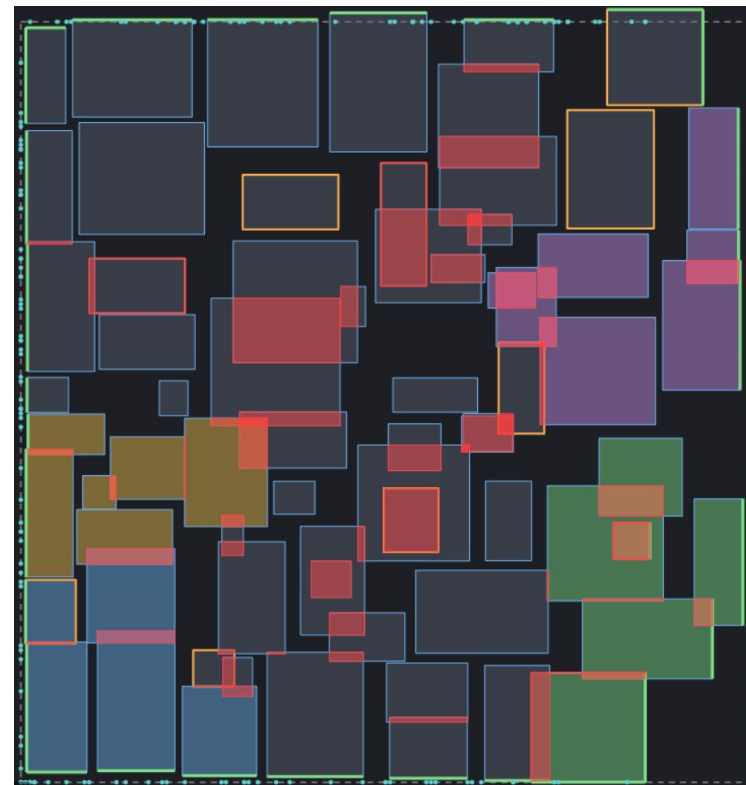
Attention Model - Result (70 blocks)

epoch 30



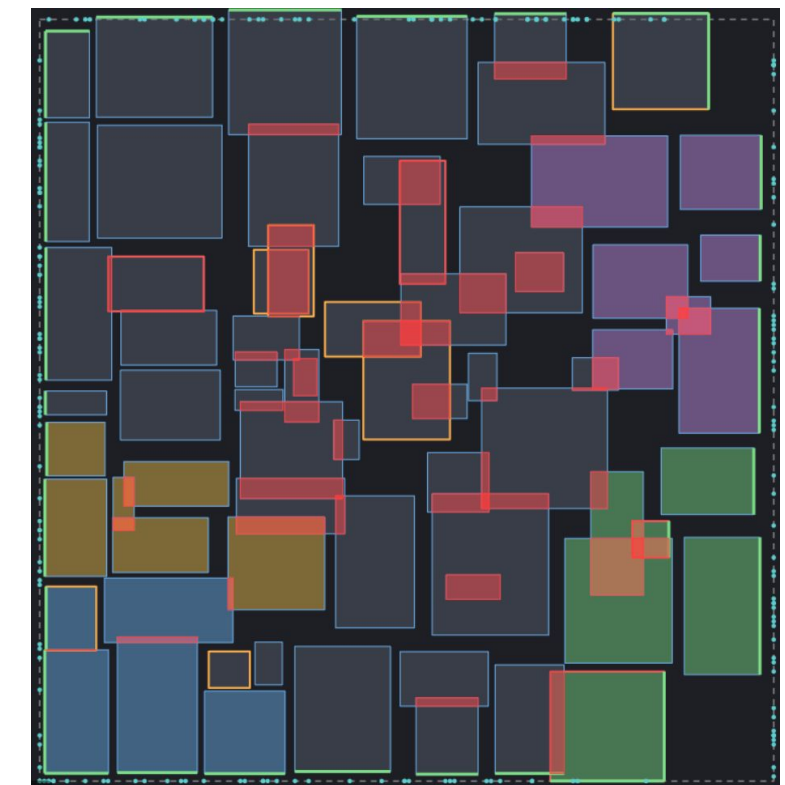
Area gap vs. optimal: +37.4%
HPWL gap vs. optimal: +46.3%
Overlap area: 5635.6 / 30538.1
Soft constraints violation: 0.813
(b = 19, g = 20, m = 0)

epoch 60

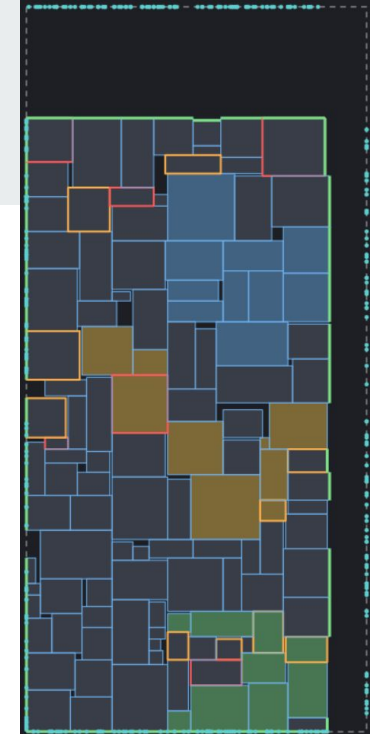


Area gap vs. optimal: +20.2%
HPWL gap vs. optimal: +30.2%
Overlap area: 2217.4 / 26704.1
Soft constraints violation: 0.625
(b = 10, g = 20, m = 0)

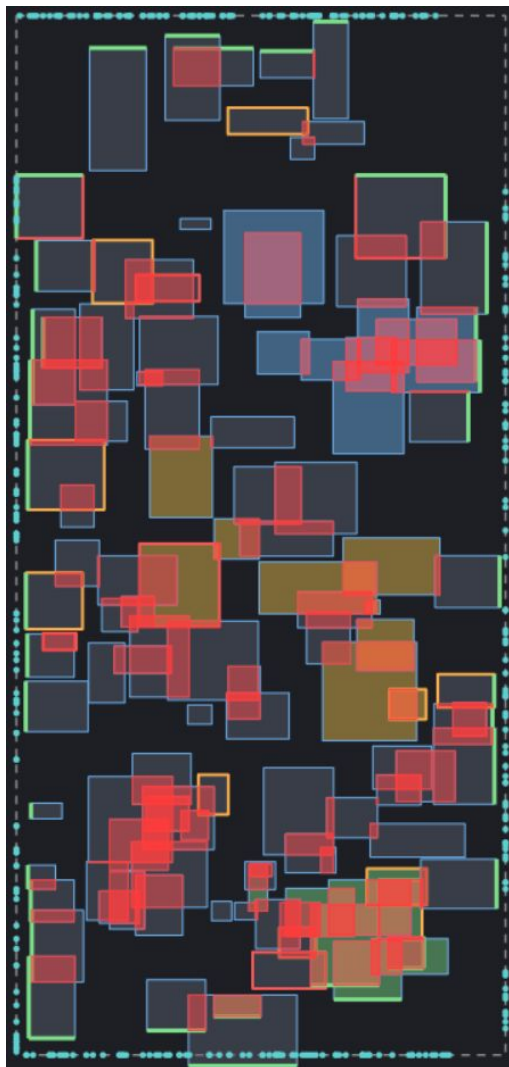
epoch 100



Area gap vs. optimal: +19.6%
HPWL gap vs. optimal: +19.2%
Overlap area: 1905.9 / 26569.6
Soft constraints violation: 0.563
(b = 7, g = 20, m = 0)

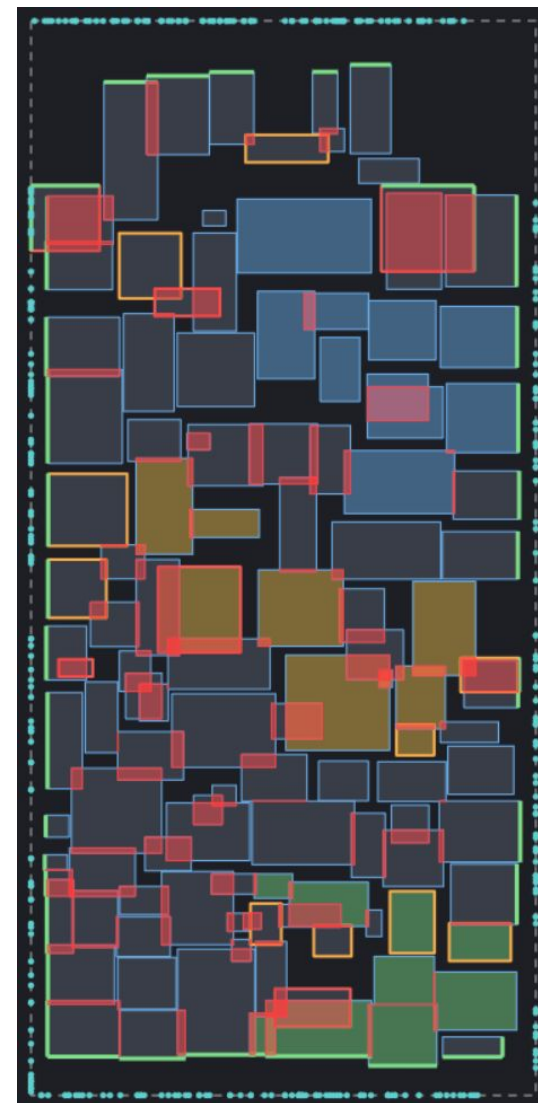


Attention Model - Result (120 blocks)



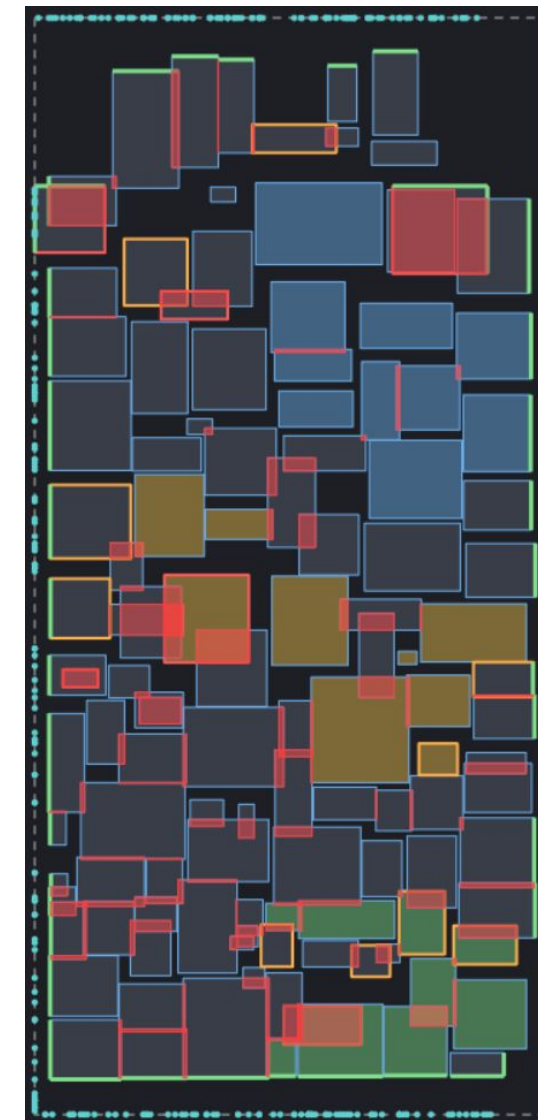
epoch 30

Area gap vs. optimal: +560.9%
HPWL gap vs. optimal: +14.7%
Overlap area: 9723.1 / 229421.8
Soft constraints violation: 0.761
(b = 26, g = 25, m = 0)



epoch 60

Area gap vs. optimal: +19.7%
HPWL gap vs. optimal: +4.0%
Overlap area: 3647.1 / 41558.3
Soft constraints violation: 0.687
(b = 21, g = 25, m = 0)



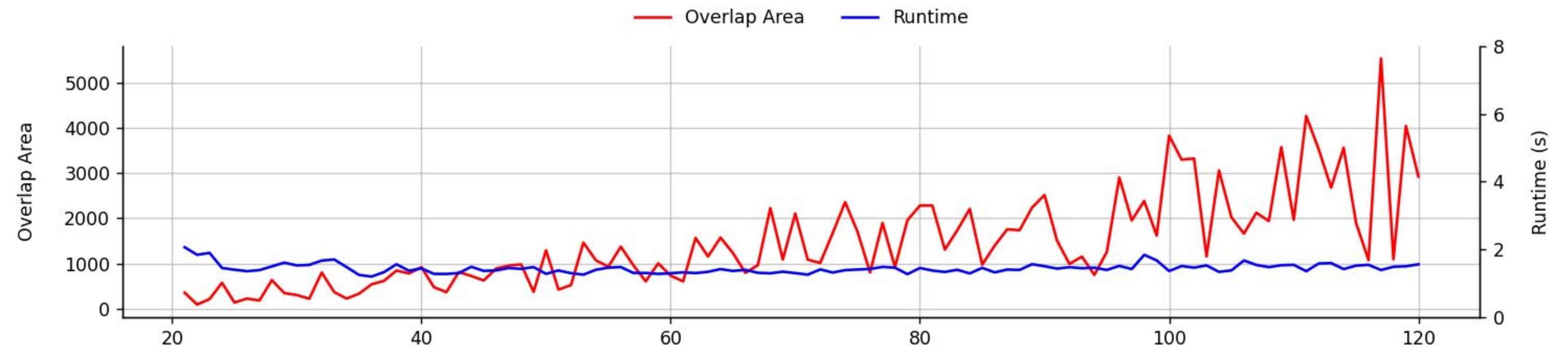
epoch 100

Area gap vs. optimal: +20.6%
HPWL gap vs. optimal: +4.0%
Overlap area: 2948.7 / 41882.8
Soft constraints violation: 0.657
(b = 19, g = 25, m = 0)

Attention Model - Metrics

Average Metrics for the Validation Set (21~120 blocks)

Avg. Chip Area Ratio	Avg. HWPL Ratio	Avg. Overlap	Avg. Soft Violation (0~1)	Avg. Runtime(s)
×1.19	×1.16	1455.48	0.5089	1.43





GNN Model - The Overlap Improvement



- Avg. Overlap: 647.9
- Accounts for soft constraints
- 270k samples @ 240 epoch
- Same model for v1.0 ~ v1.2



- Avg. Overlap: 518.8 (-20%)
- Sweep for the optimal inference profile



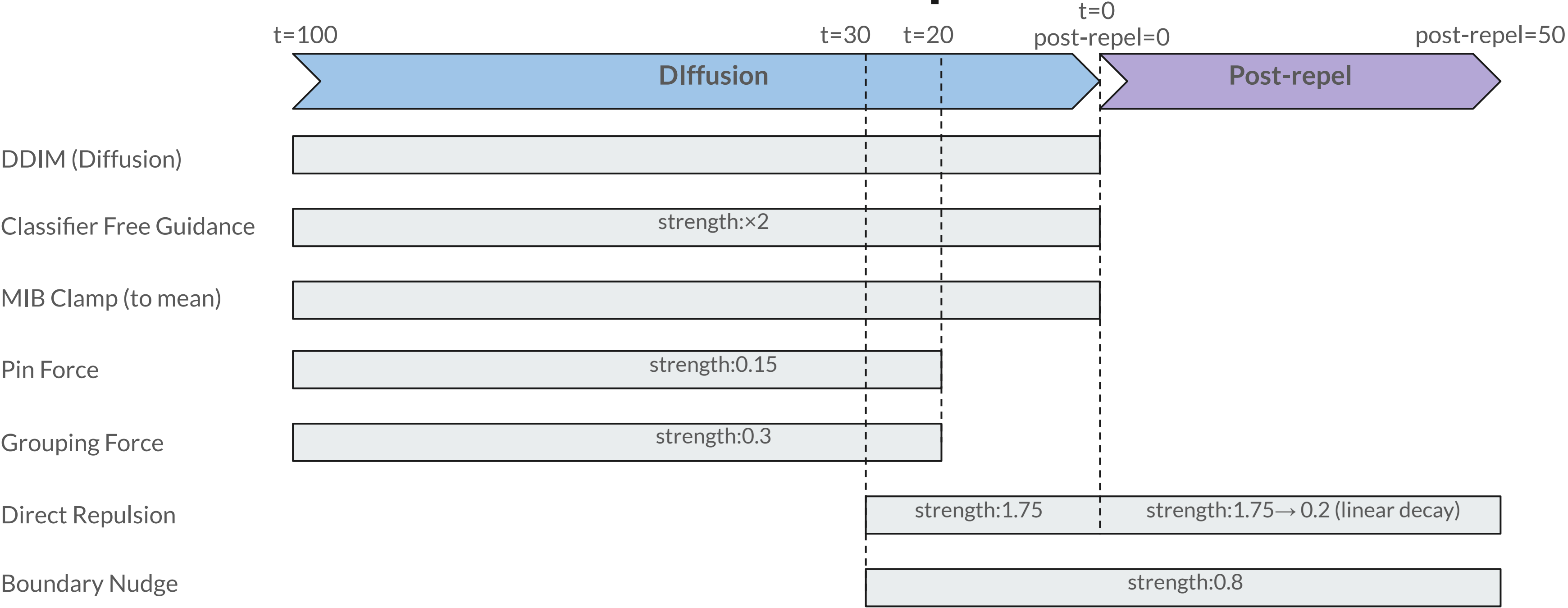
- Avg. Overlap: 254.6 (-51%)
- DDIM 50 -> 100
- Re-noise refinement
- v1.2.1 adds post-repel, with
Avg. Overlap: 55.6 (-78%)

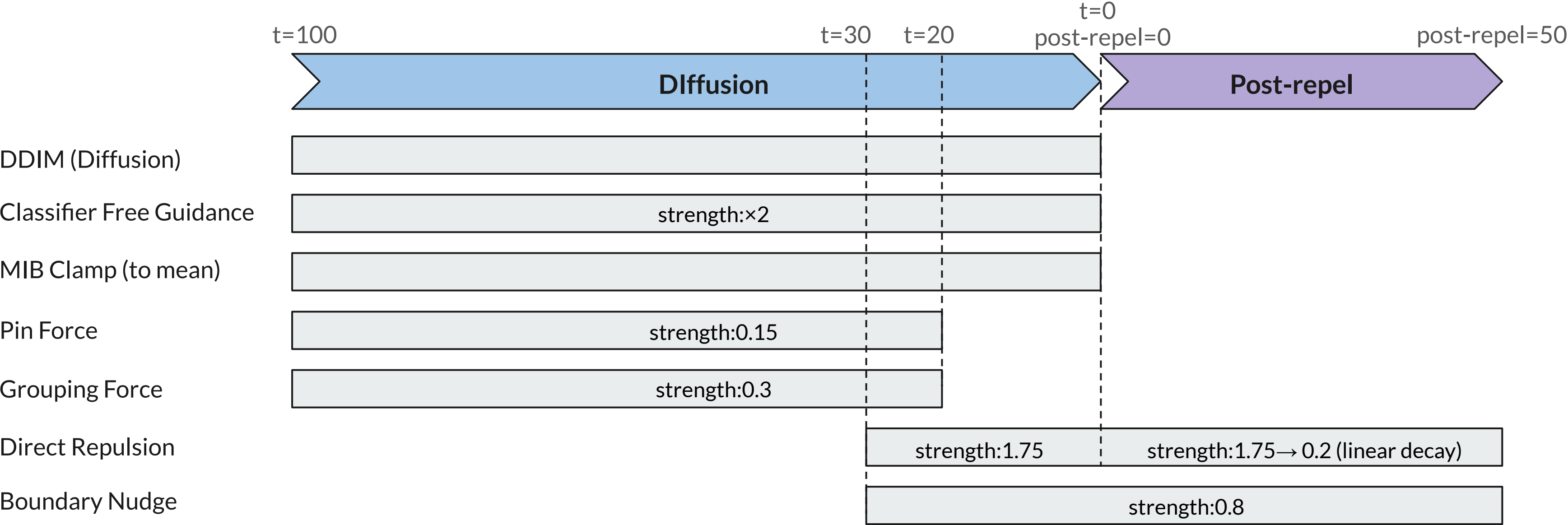


GNN Model - Training

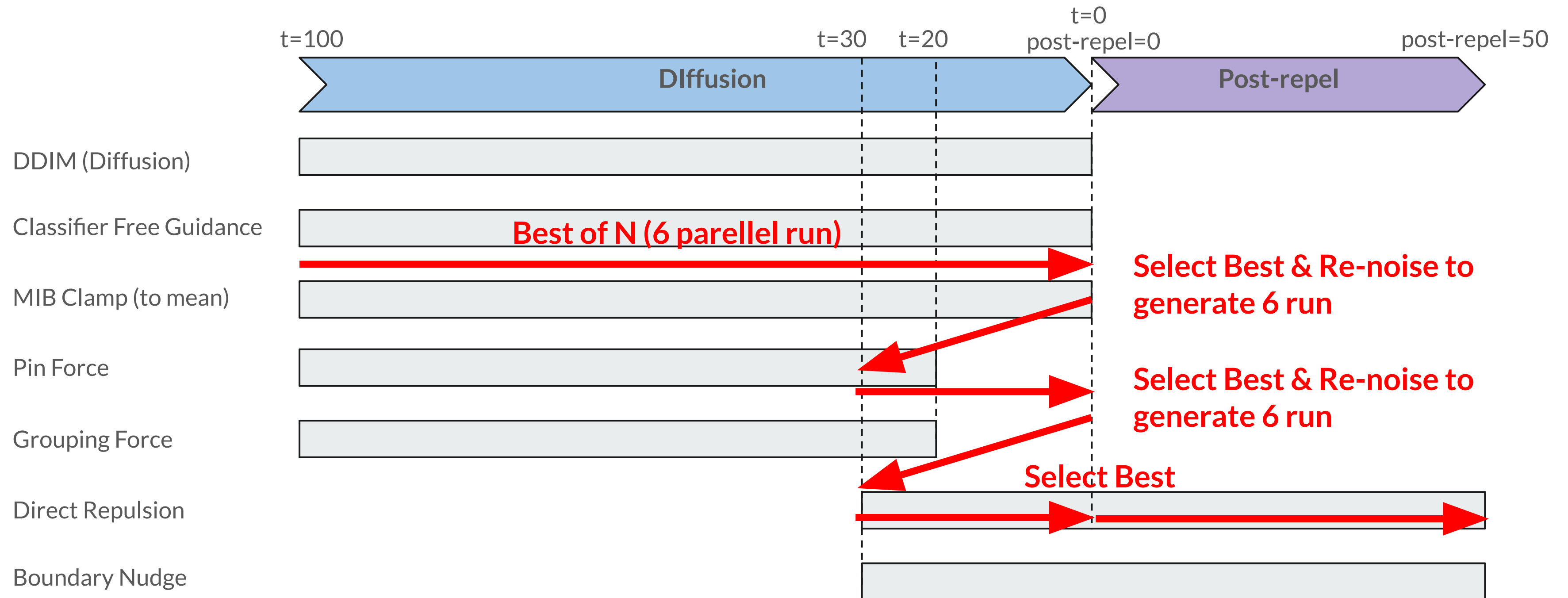
- **Loss Functions**
 - Standard MSE: Block position, block aspect-ratio
 - Auxiliary Loss Functions: Overlap loss, Pin attraction loss
- Pin as Block Attribute: Feed the centroid position of connected pins as block attribute.
- KNN-edges: Construct additional edges to K closest blocks.
- Even more: Edge pruning, data augmentation

GNN Model - Inference Techniques



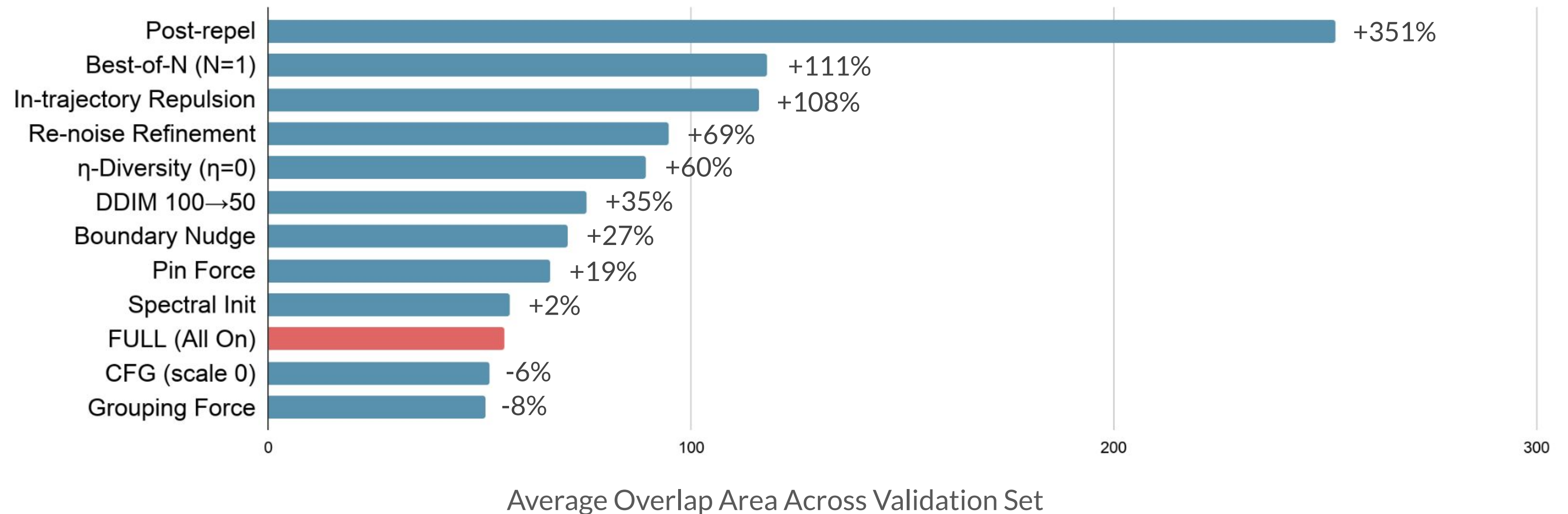


GNN Model - Best of N & Re-noise



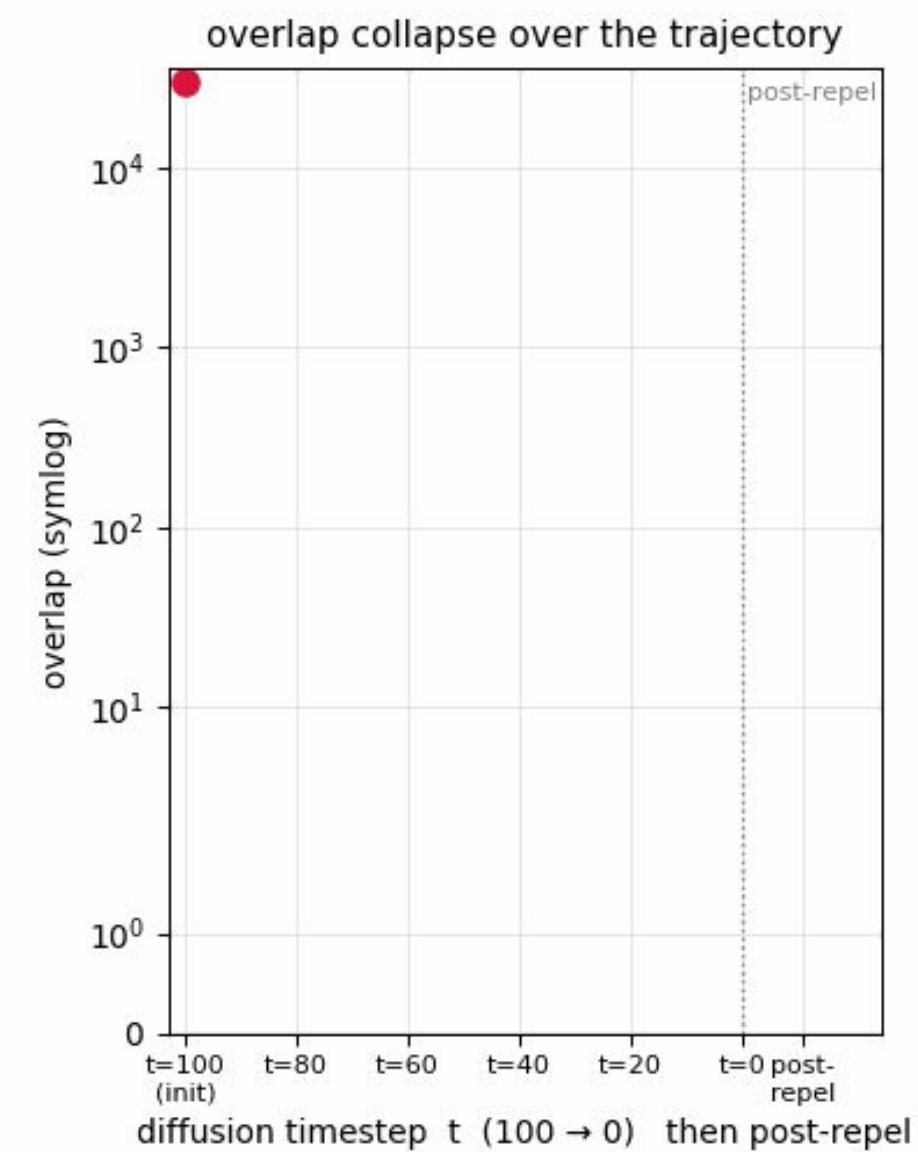
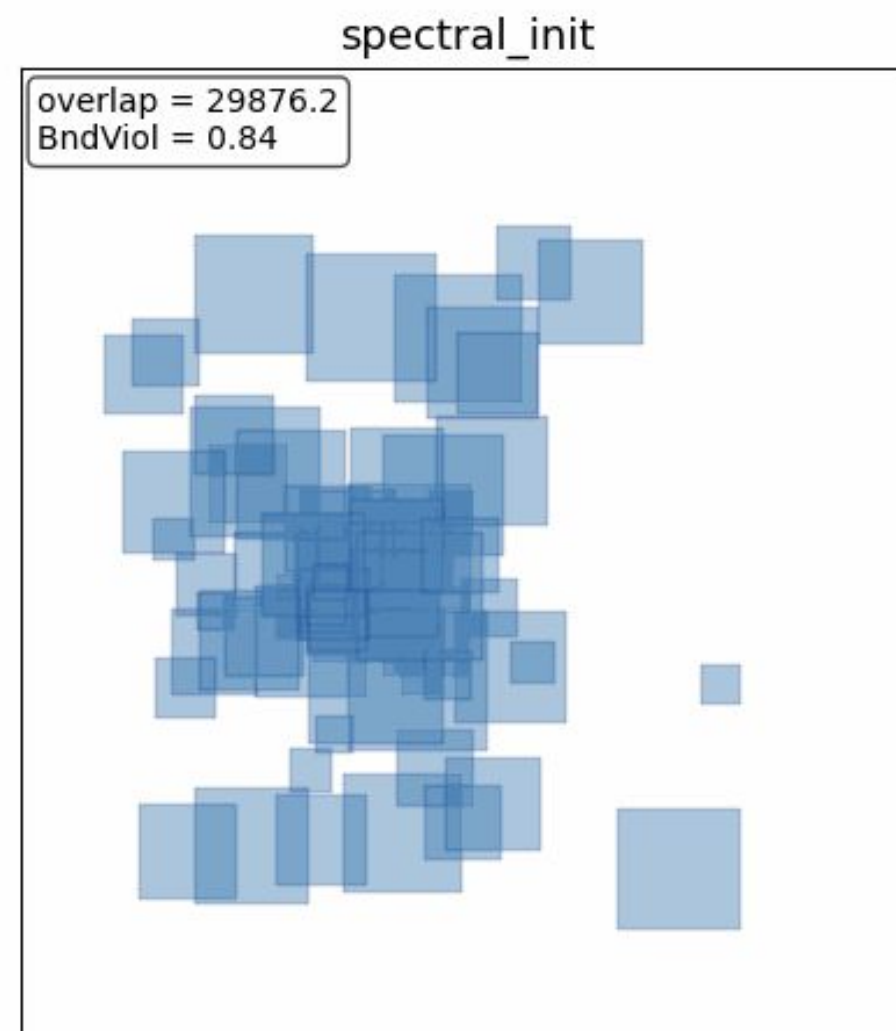


GNN Model - Ablation Test on the Inference Stack



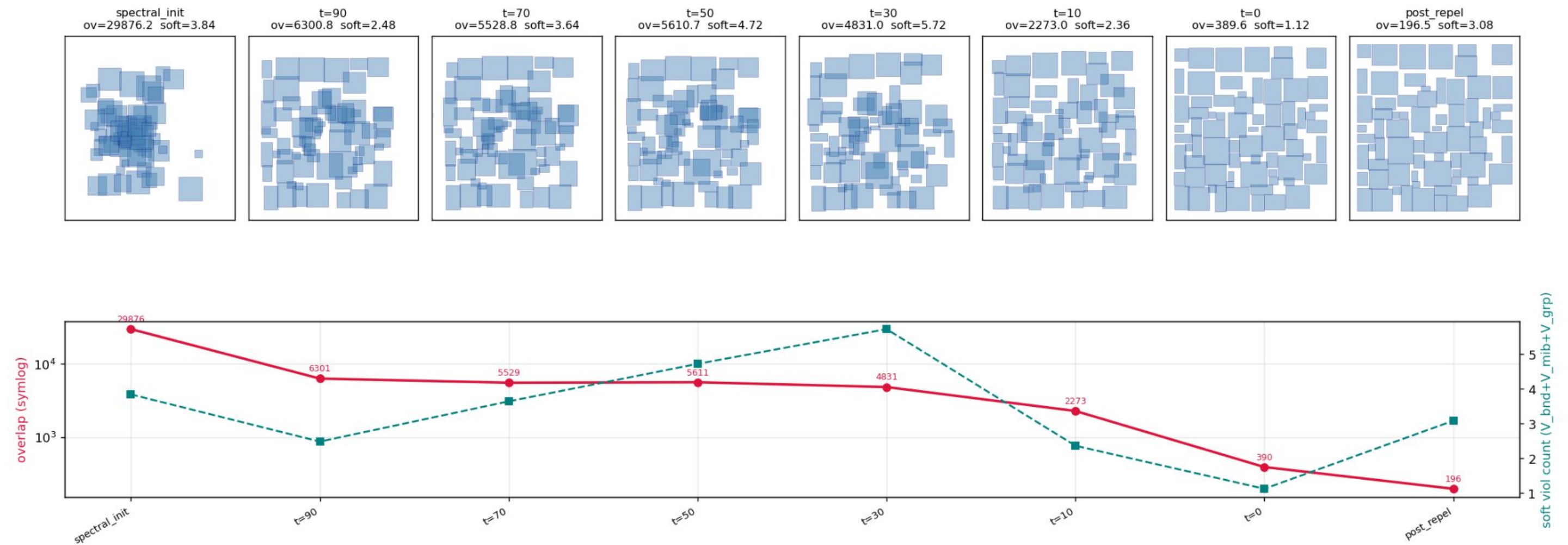
GNN Model - Visualization

Inference animation — official 70-block sample (ddim100, $\eta=0.5$, post_repel50)

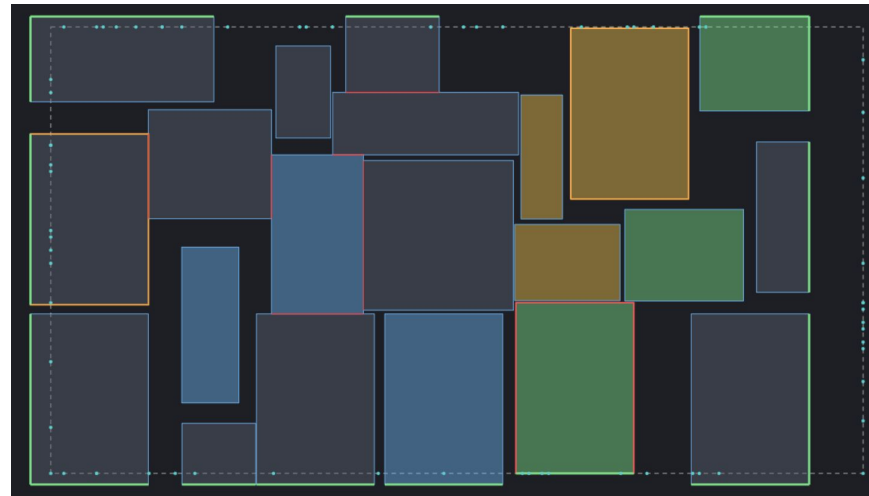


GNN Model - Visualization

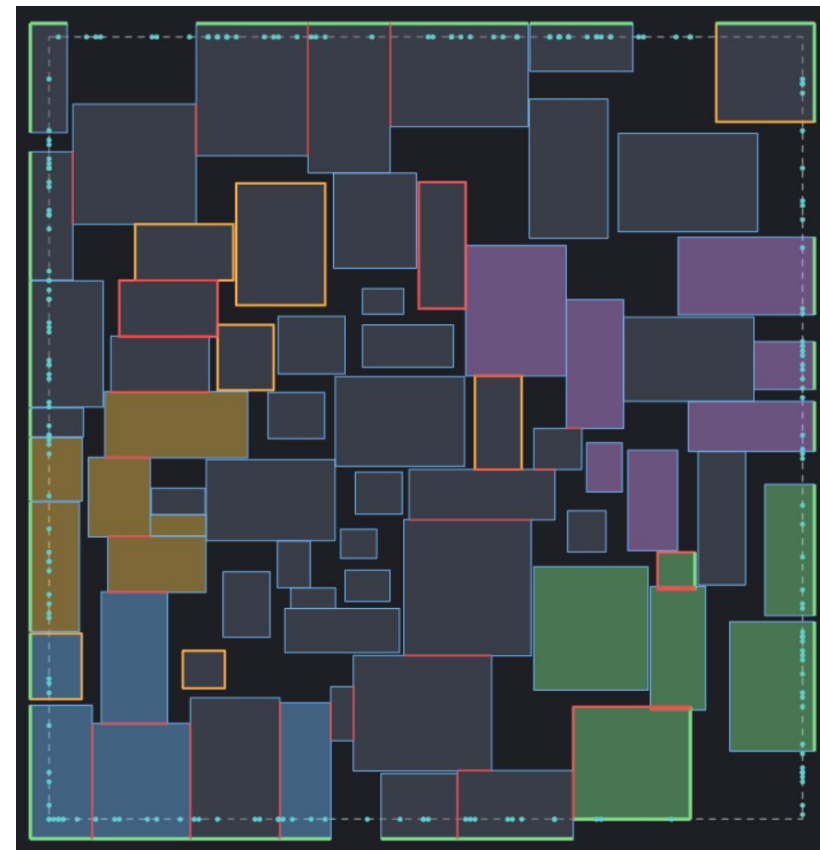
Inference trace — official 70-block sample
v1.2.1 | ddim_steps=100, eta=0.5, refine=2, repel=1.75, post_repel=50(+bnd)



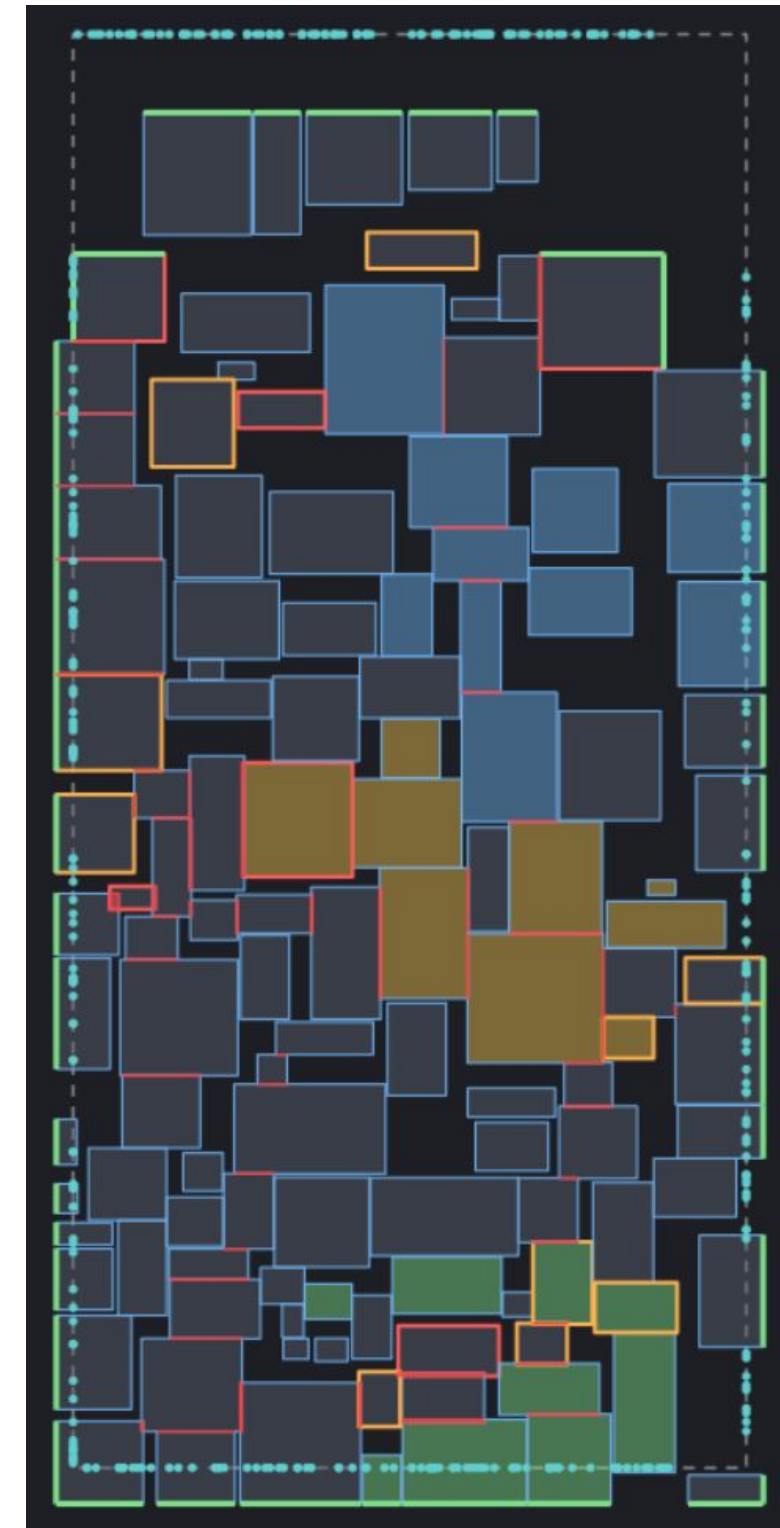
GNN Model - Result



21 Blocks



70 Blocks

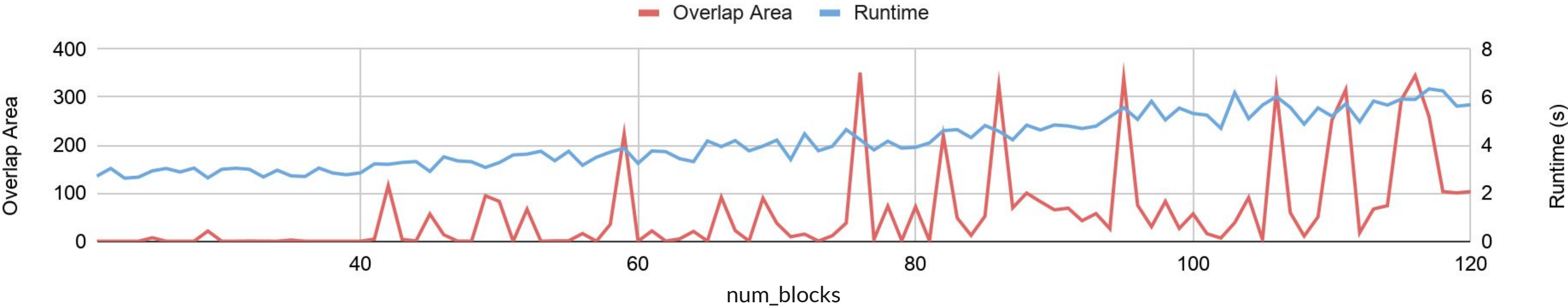


120 Blocks

GNN Model - Metrics

Average Metrics for the Validation Set (21~120 blocks)

Avg. Chip Area Ratio	Avg. HWPL Ratio	Avg. Overlap	Avg. Soft Violation (0~1)	Avg. Runtime(s)
×1.3	×1.3	55.6	0.603 / 0.145 (1% snapped)	4.13





Comparison

Average Metrics for the Validation Set (21~120 blocks)

	Avg. Chip Area Ratio	Avg. HWPL Ratio	Avg. Overlap	Avg. Soft Violation (0~1)	Avg. Runtime(s)
Attention Model	×1.19	×1.16	1455.48	0.5089	1.43
GNN Model	×1.3	×1.3	55.6	0.603 / 0.145 (1% snapped)	4.13



Legalization: 核心 Pipeline (3 個 Stage)

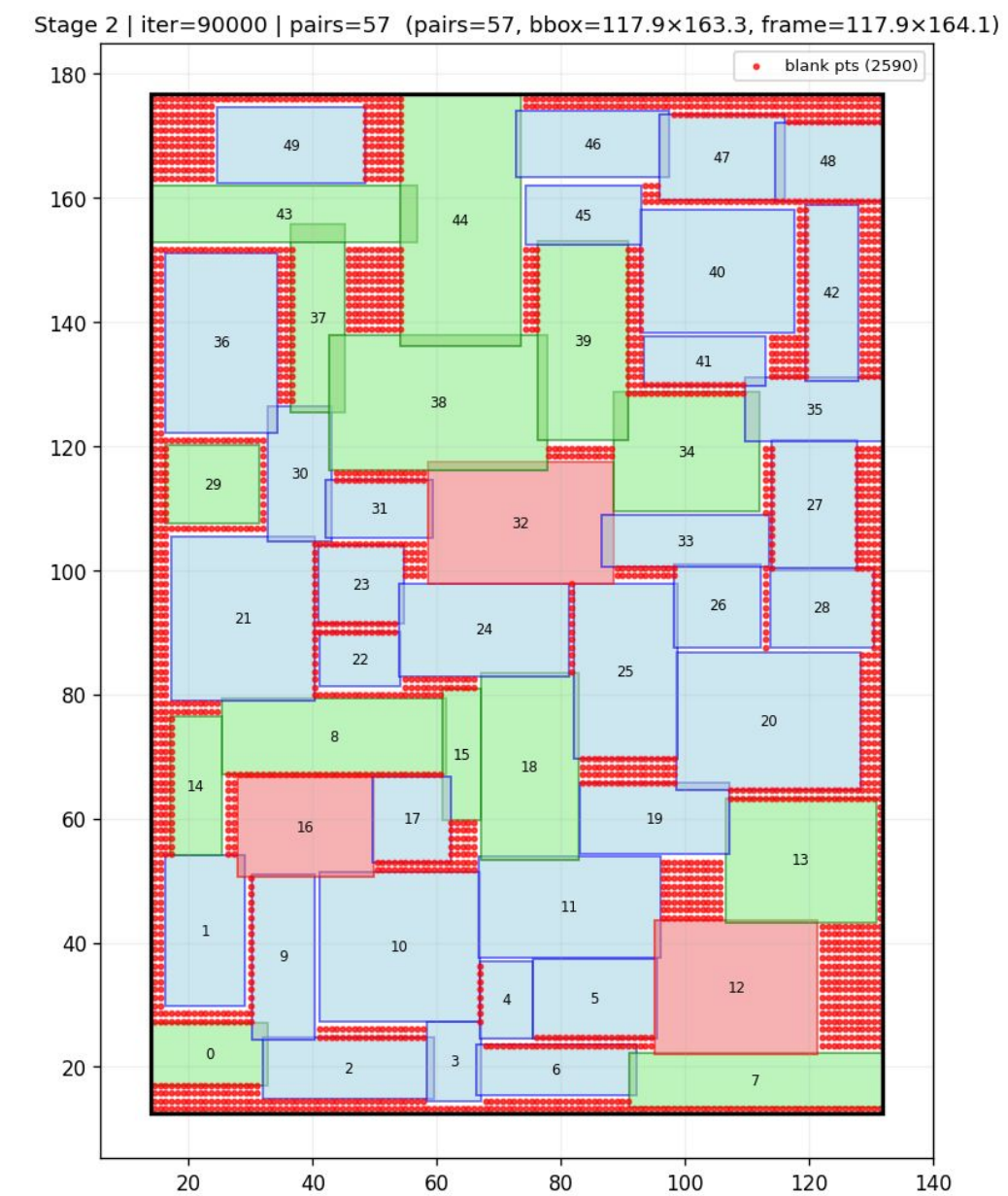
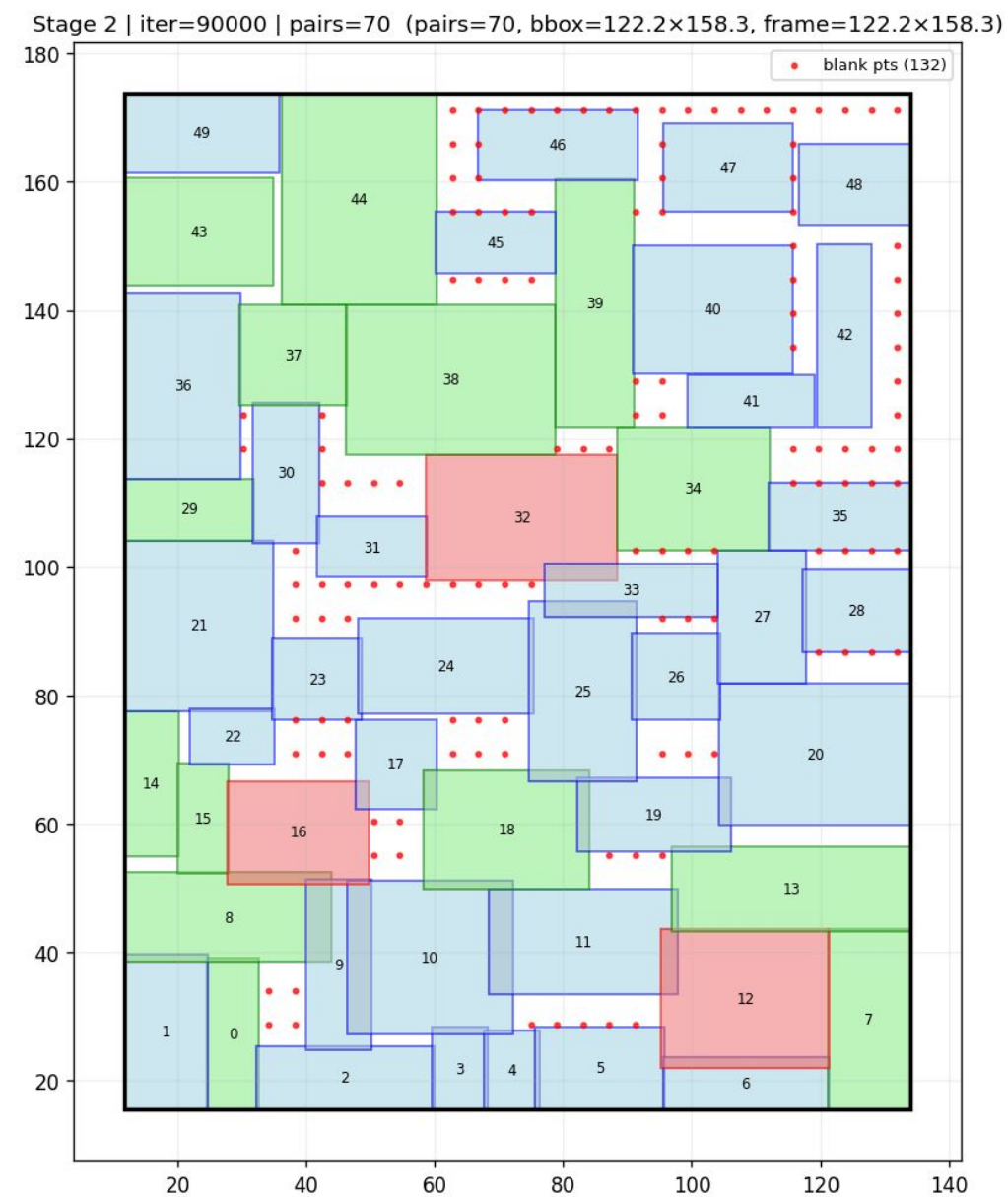
- Stage 1 (框收縮): 跑 T_{max} 次迭代的力導向演算法, 5 種力同時作用:
- Stage 2 (框凍結細修): 再跑一次 T_{max} , 框不再縮小, α 重置後重新衰減, 額外 啟用:
- Stage 3 (Finalize): 左下 → 右上掃描, 逐塊切齊重排保證零重疊。目前暫時關閉, 只做 `align_to_origin`



Legalization: 核心 Pipeline (3 個 Stage)

- Stage 1 (框收縮): 跑 $T_{\max}$ 次迭代的力導向演算法, 5 種力同時作用:
 - 重疊排斥力 (k_{rep}) 把兩兩重疊的 block 沿較淺軸推開
 - Anchor spring (k_{anchor}) 保留 diffusion 輸出的原始位置結構
 - AR bias 防止布局被壓成長條形
 - Shrink frame 把虛擬矩形框從初始 bbox 線性收縮到目標大小 ($\text{block 面積} \times 1.15$)
 - Blank-point grid 用 128^2 採樣點吸引 block 填補空洞

Legalization: 核心 Pipeline(3 個 Stage)

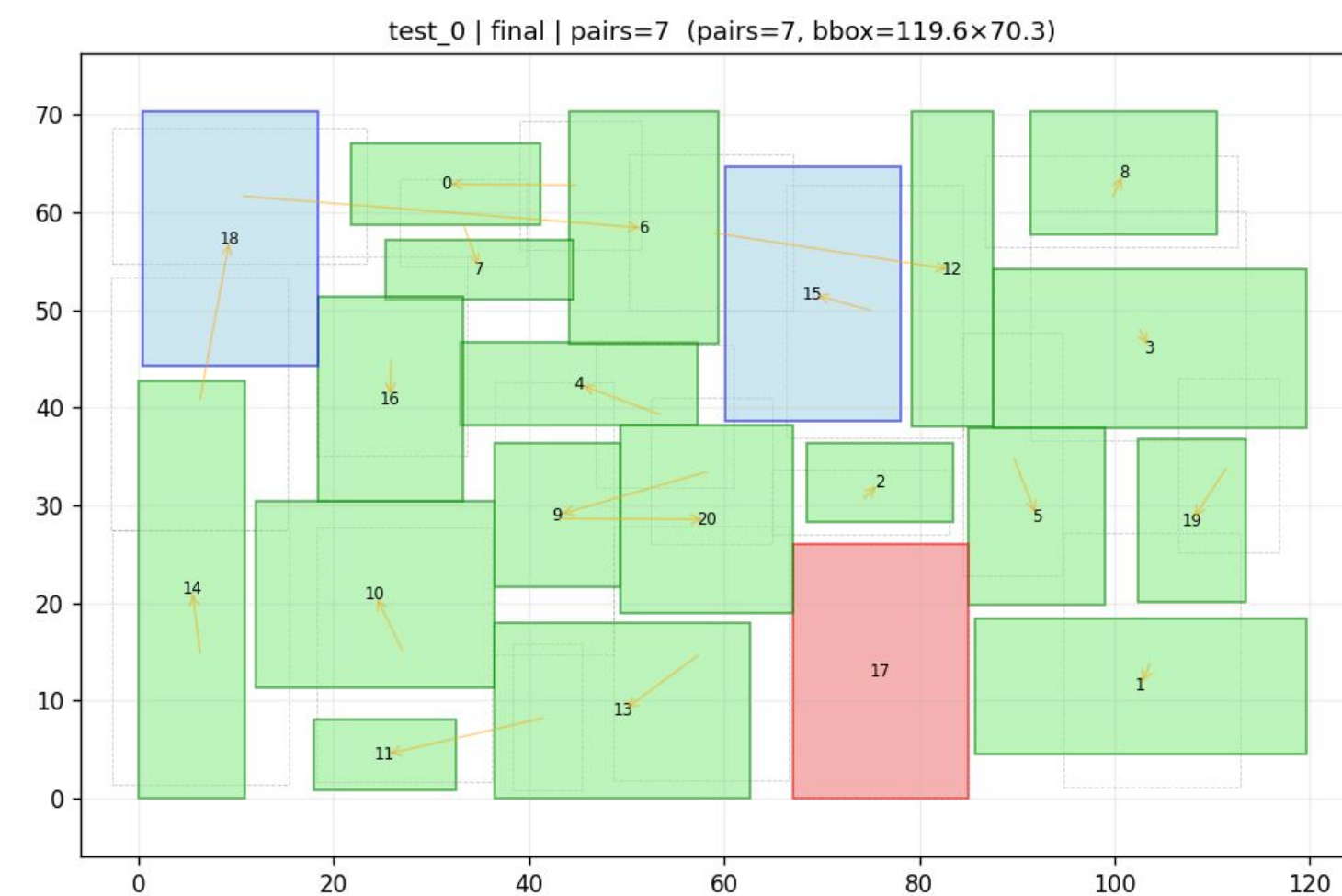
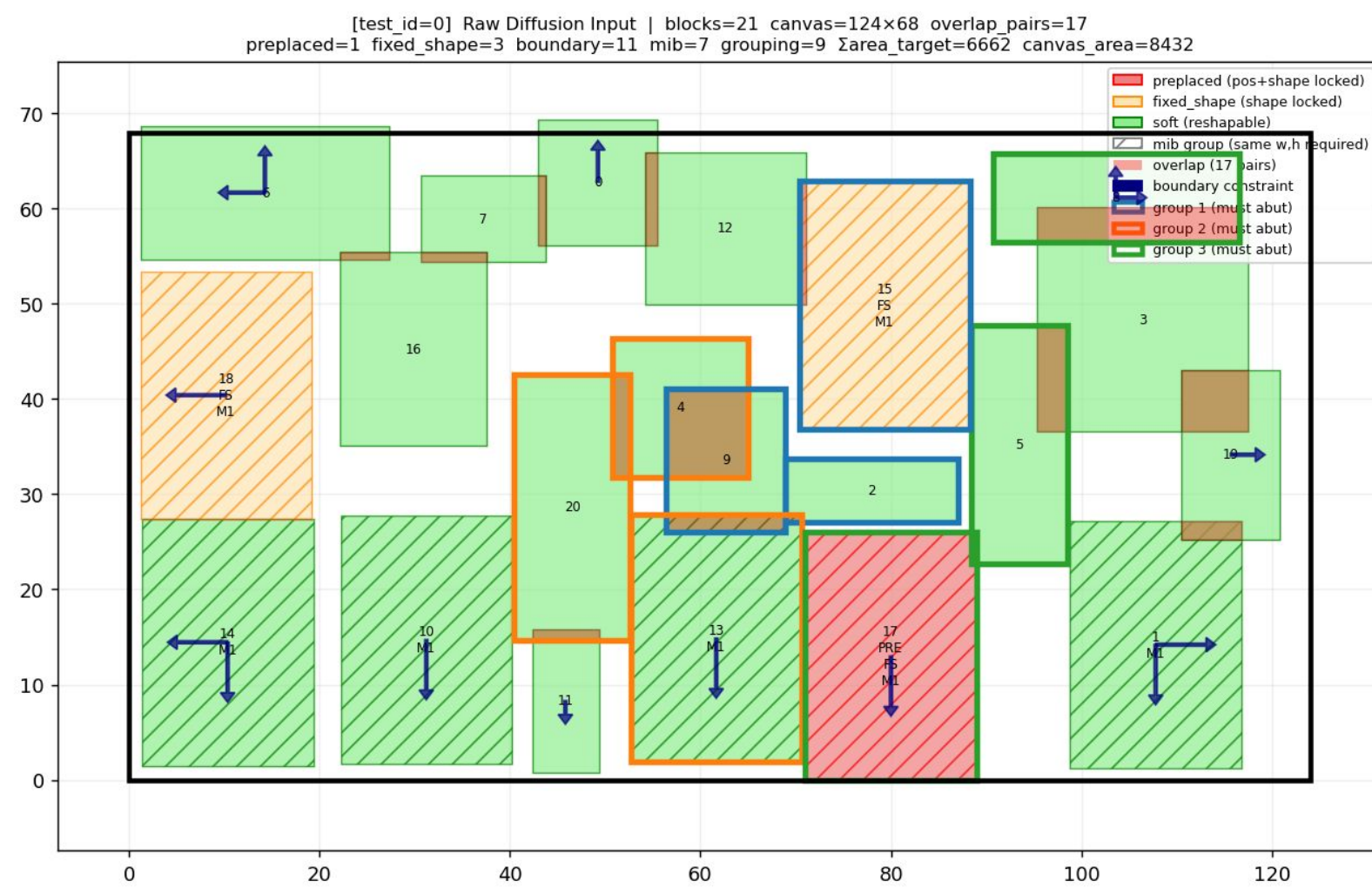




Legalization: 核心 Pipeline (3 個 Stage)

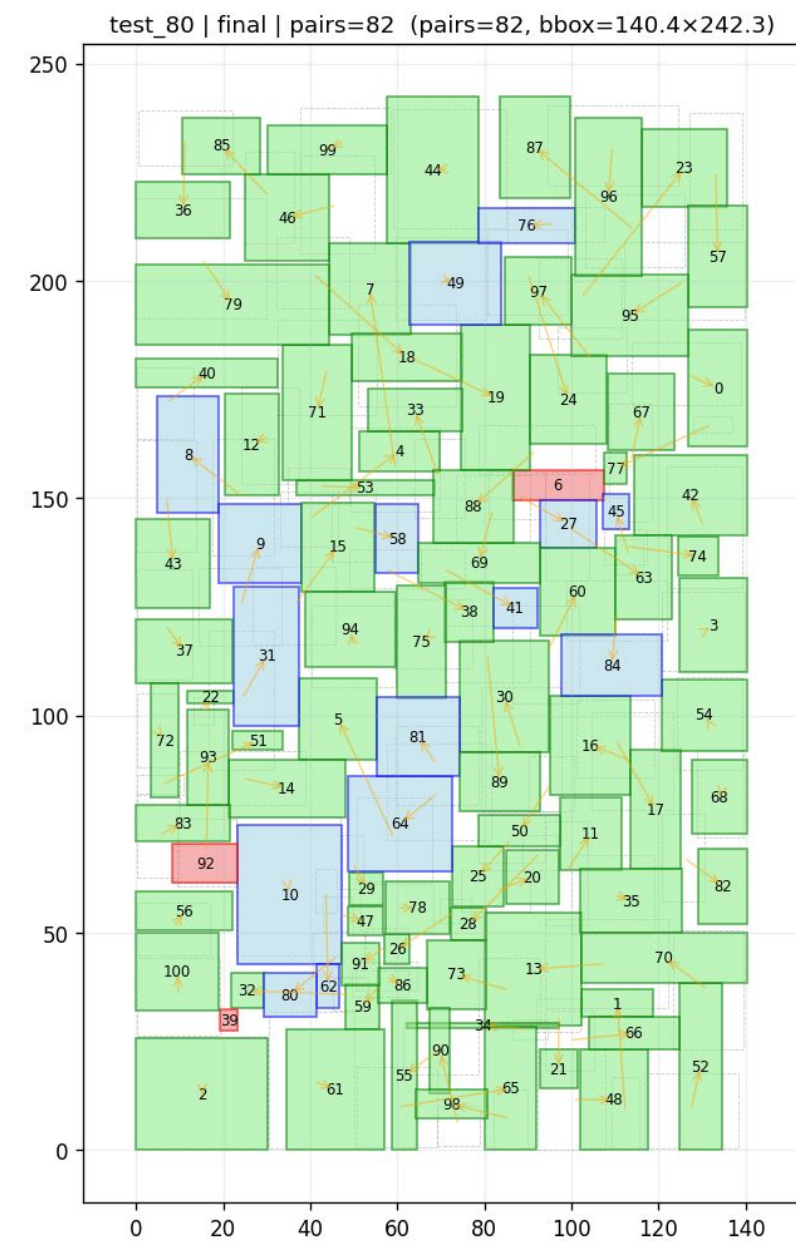
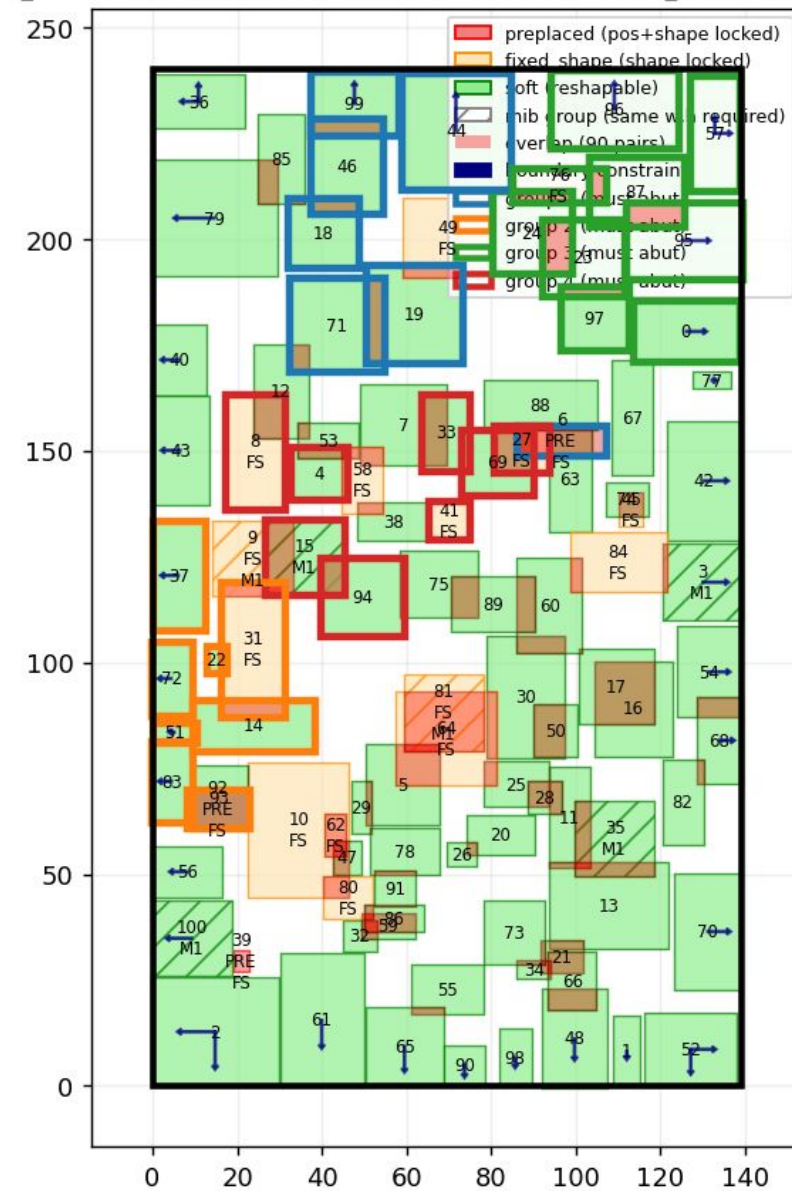
- Stage 2 (框凍結細修): 再跑一次 T_{max} , 框不再縮小, alpha 重置後重新衰減, 額外 啟用:
 - Soft block 面積守恒 reshape (依左右 / 上下壓力比例調整長寬比)
 - Lift-and-replace (每 10% 進度把最擁擠的 block 搬到附近最空曠處)
- Stage 3 (Finalize) (debug 中): 左下 → 右上掃描, 逐塊切齊重排保證零重疊。目前暫時關閉, 只做 align_to_origin。

Legalization: Results (Attention)

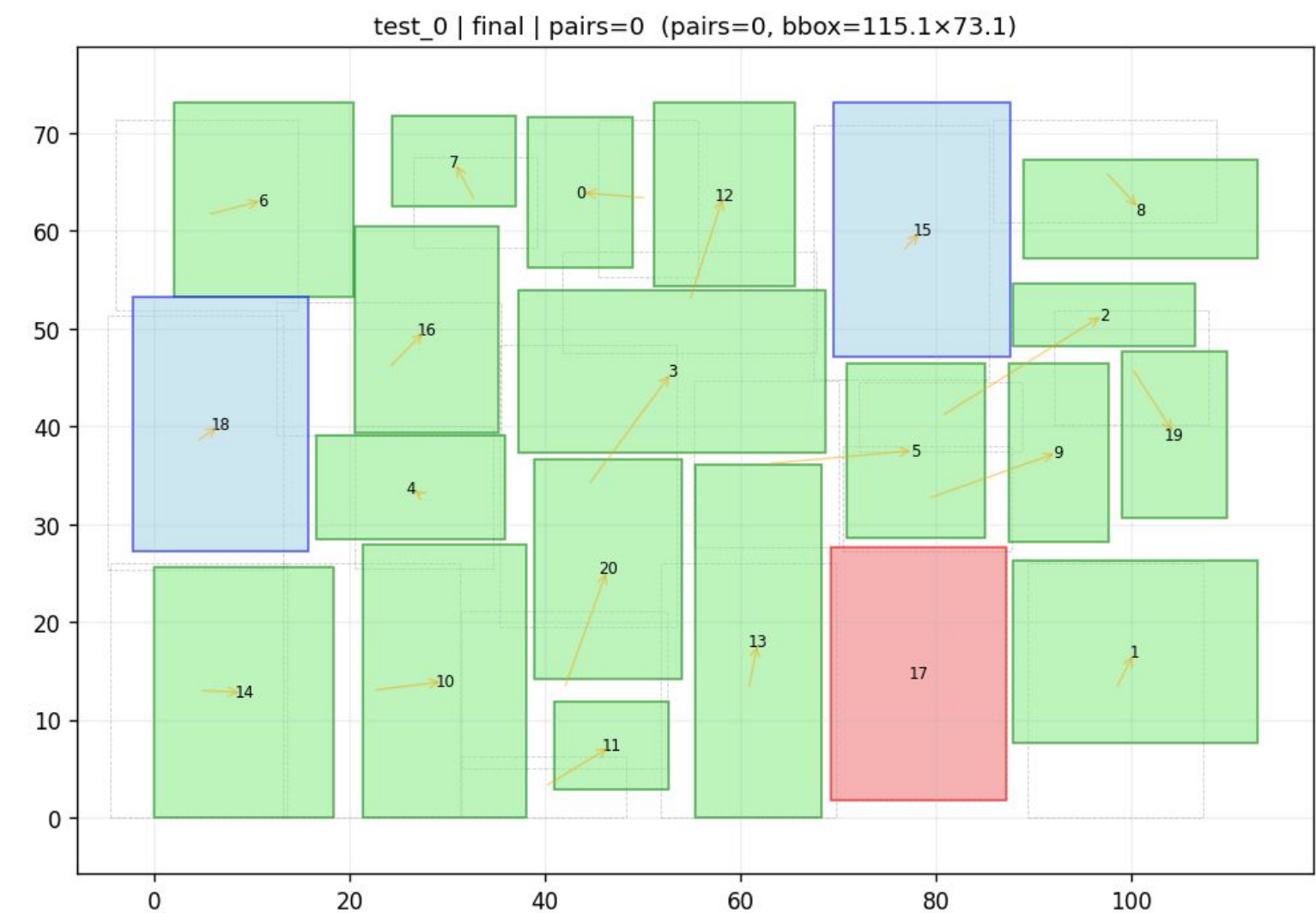
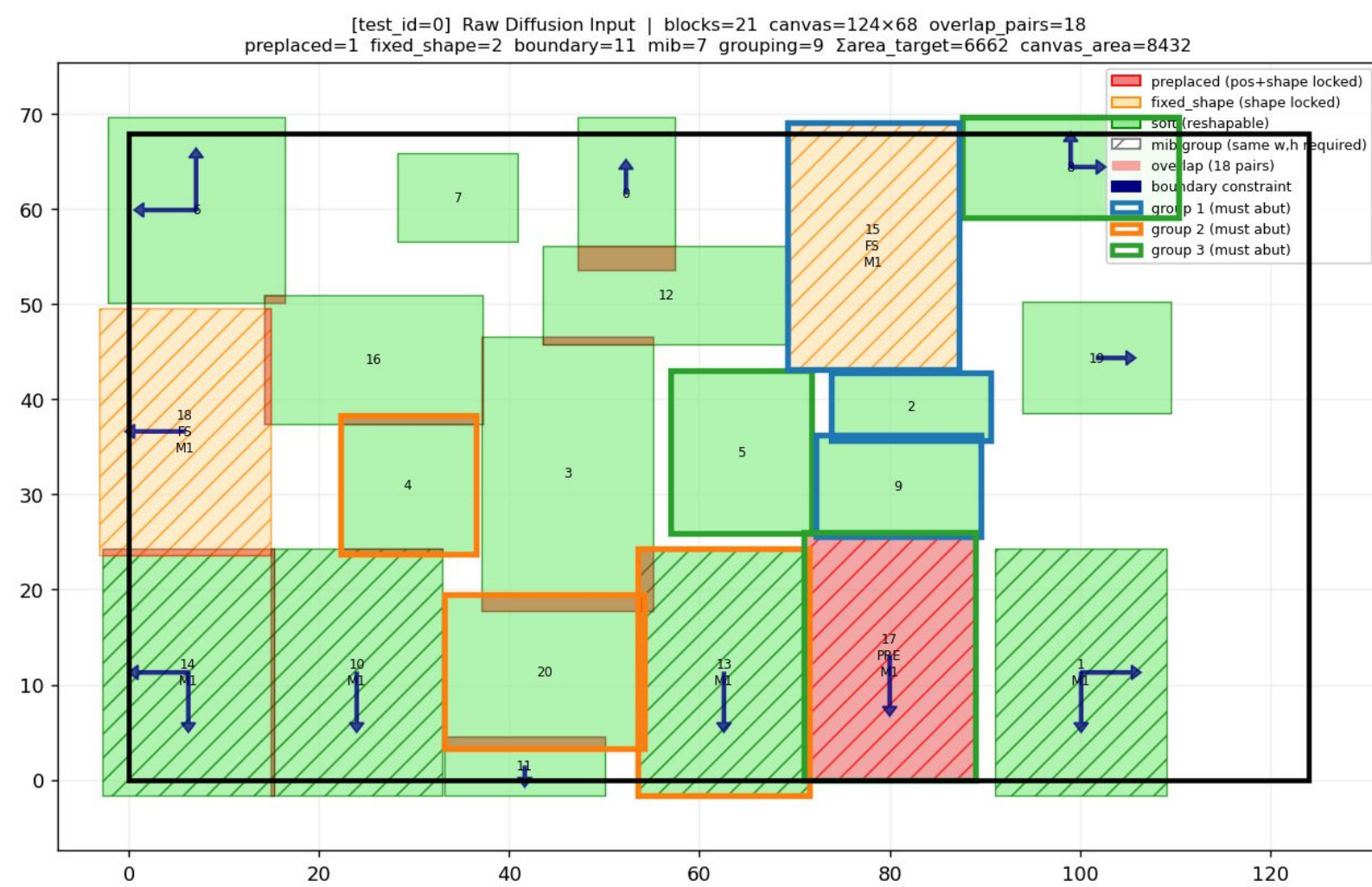


Legalization: Results (Attention)

[test_id=80] Raw Diffusion Input | blocks=101 canvas=139×240 overlap_pairs=90
preplaced=3 fixed_shape=18 boundary=30 mib=6 grouping=32 Σ area_target=27680 canvas_area=33360

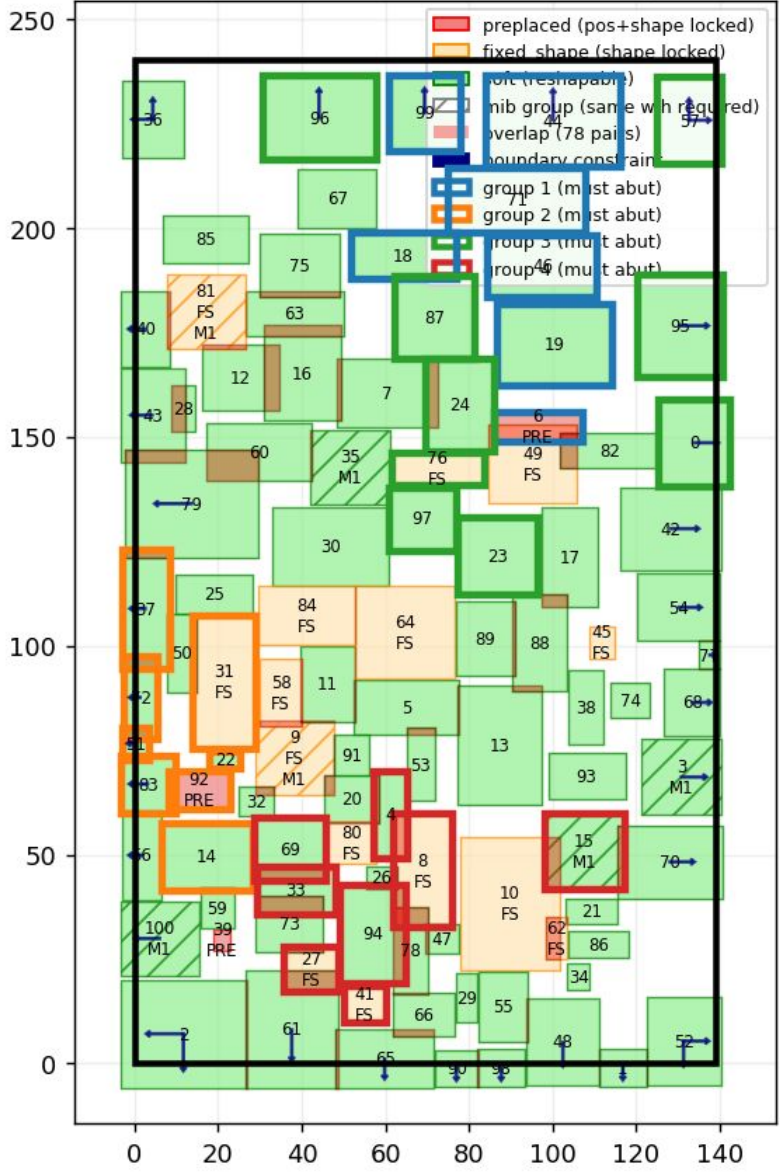


Legalization: Results (GNN)

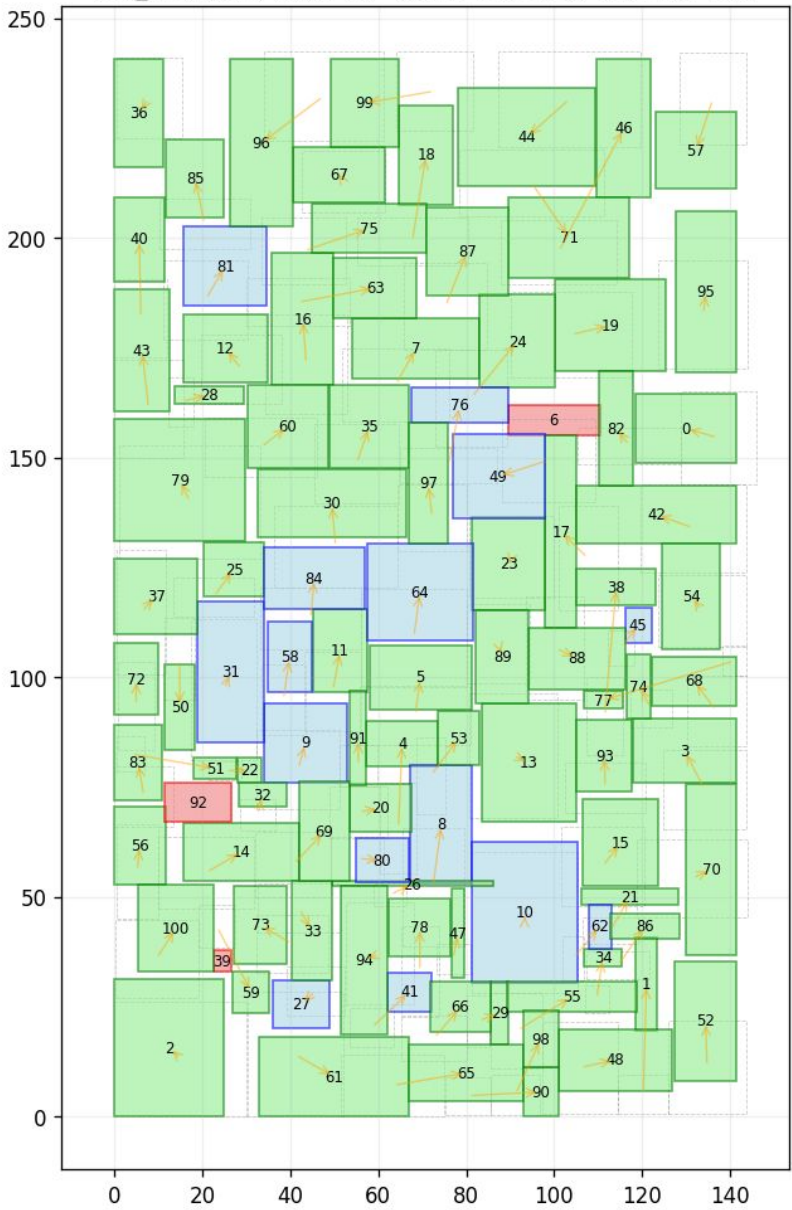


Legalization: Results (GNN)

[test_id=80] Raw Diffusion Input | blocks=101 canvas=139x240 overlap_pairs=78
preplaced=3 fixed_shape=15 boundary=30 mib=6 grouping=32 Σ area_target=27680 canvas_area=33360



test_80 | final | pairs=67 (pairs=67, bbox=141.4x240.6)





Legalization: Conclusion

- Force-directed core (Stage 1+2) fully implemented — repulsion, anchor spring, shrink frame, blank-point attraction, soft reshape, lift-and-replace all working
- Dual-format input supported — CSV and Diffusion JSON (GNN + Attention)
- Batch execution & metrics — run_all.py, run_diffusion.py, full metrics pipeline ready
- Time: 2 ~ 10 min/case



Legalization: Conclusion & Unfinished Task

- Stage 3 (Finalize) disabled — code exists in `finalize.py` but fully commented out in `main.py`; no zero-overlap guarantee on output
- No cleanup pass — residual 50–100 overlap pairs remain after Stage 1+2 with no post-processing
- Boundary constraints unused — boundary (edge-snapping), mib (same w/h group), grouping (must-abut) from JSON are parsed but never enforced
- Hard/soft swap experimental — `enable_hard_soft_swap=False` by default, stability unverified